# From Anonymous Shapes to Named Places

## A Tool for Braille and Place-Semantic Annotation of Tactile Maps

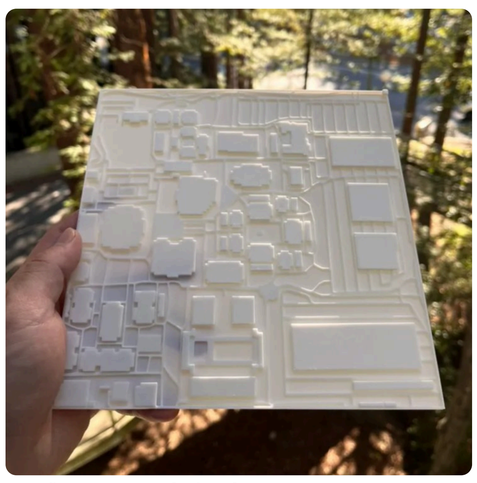
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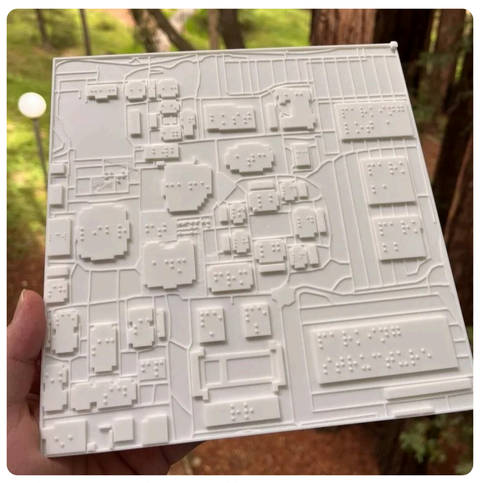
Try the tool

## On a tactile map, a building under the finger is an anonymous shape.

Our tool adds Braille labels to tactile maps that already exist, so each shape on the print carries its own name.



**Before** · no labels



**After** · with Braille




## Why we need it A tactile map gives the *where*. The *what* is hard to get.

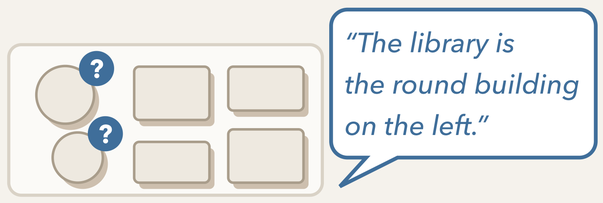
1



### Felt, but not named

On a Touch Mapper† print, every building is a shape with no name.

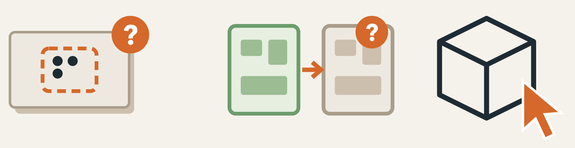
2




### Described, but hard to match

A spoken description is hard to tie to the right shape.

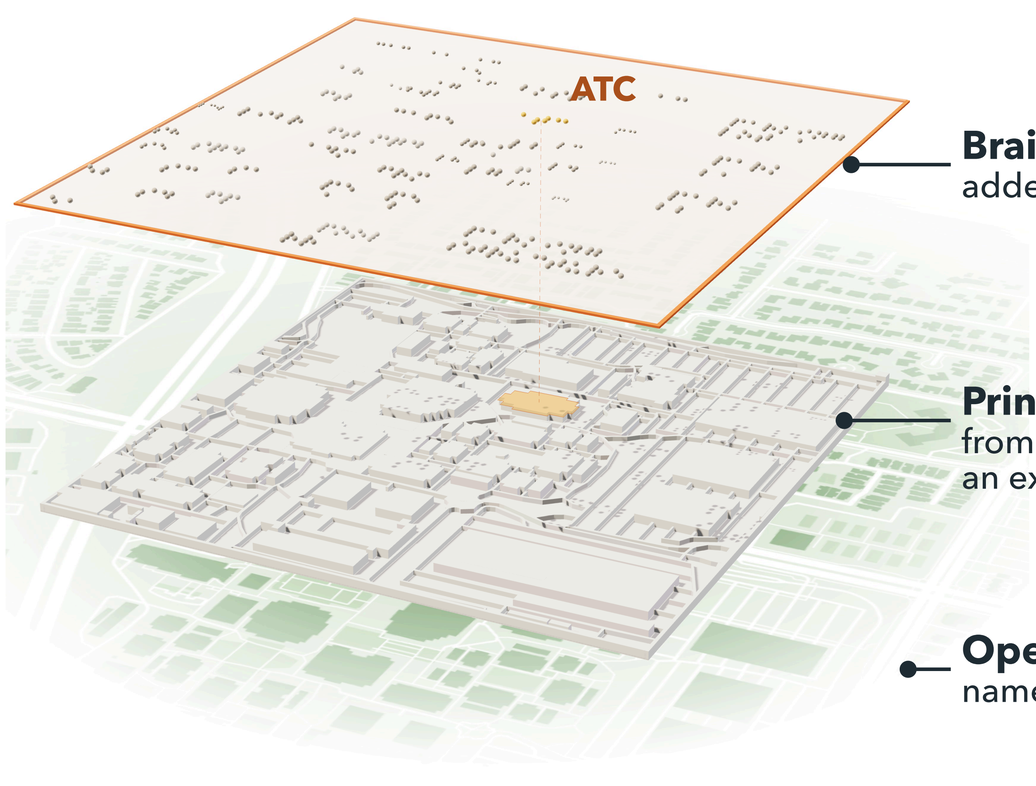
3




### Hard to label by hand

Each label means matching the print to a map, then editing it in 3D software.

## What our tool does Click to annotate. Easy to locate, edit and customize.

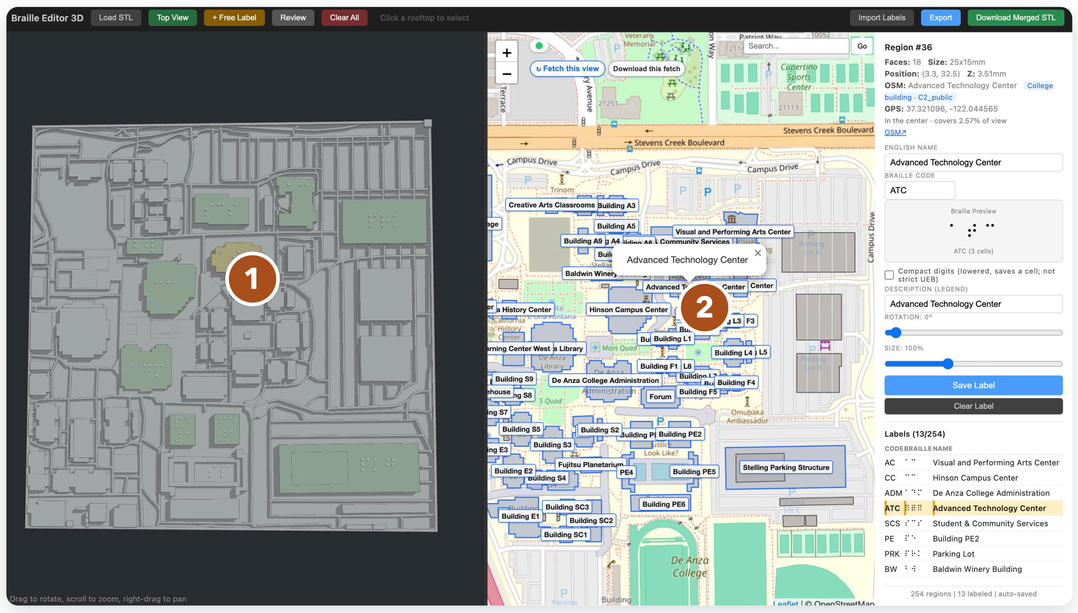



**Braille layer**
added with our tool

**Printed base**
from Touch Mapper, an existing tool

**OpenStreetMap**
names and places

**3D model** from Touch Mapper | **OpenStreetMap** | **Label**




1 Load the Touch Mapper model, click a building.

2 Click it on OpenStreetMap. Name and Braille fill in.

3 Review, then export. The Braille attaches to the model base.

## What comes next From one labeled print to a shared Braille map of the world

**Today: one print**

- Braille on the print
- Braille legend
- Print-ready 3D model

**Each label is saved**

a shape on the print ↓ a place on OpenStreetMap

**Next: a shared, reusable map**

- Reused for the next print
- Audio, symbols, AI agents
- Built together by everyone
- Embedded in OpenStreetMap

**Project page** leolee7.github.io/#project-tactile-annotation · **Paper** doi.org/10.1145/3797867.3841235

†Touch Mapper, an existing tool that makes tactile maps from OpenStreetMap: touch-mapper.org
*Work unrelated to position at Amazon.

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